THE MASTERMIND COACHING CLASSES

Revision Notes

Class 7 · Mathematics

Ganita Prakash — Part 1, Chapters 1 to 4 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	Large Numbers Around Us	Place value to crores, Indian and international systems, estimation, large-number problems
2	Arithmetic Expressions	Order of operations, brackets, forming expressions, properties and shortcuts, expression puzzles
3	A Peek Beyond the Point	Decimal place value, comparing, rounding, units, addition, subtraction, multiplication and division
4	Expressions Using Letter-Numbers	Letter-numbers, algebraic expressions, terms and coefficients, simplifying, generalising patterns

The Mastermind Coaching Classes — Road No. 5, VKIA, Jaipur — 9887258679

CHAPTER 01 · MATHS (PART 1)

Large Numbers Around Us

Populations, distances and money run into lakhs, crores and beyond. This chapter extends place value to crores and billions, compares the Indian and international ways of writing numbers, rounds and estimates large quantities, and solves real problems such as how many seconds there are in a day.

Big Questions

- How are big numbers read, written and compared in the Indian and international systems?
- How do we round and estimate large quantities?
- How do we make sense of very large numbers in everyday contexts?

1.1 Place Value Beyond Lakhs

Each place is **10 times** the place on its right. Adding 1 to the largest 6-digit number 9,99,999 gives **10,00,000 (ten lakh)**; adding 1 to 99,99,999 gives **1,00,00,000 (one crore)**.

Name	Number	In powers of 10	Words
Thousand	1,000	10^3	one thousand
Ten thousand	10,000	10^4	ten thousand
Lakh	1,00,000	10^5	one lakh = 100 thousands
Ten lakh	10,00,000	10^6	ten lakh = 1 million
Crore	1,00,00,000	10^7	one crore = 100 lakhs
Arab (hundred crore)	1,00,00,00,000	10^9	100 crore = 1 billion

1.2 Indian and International Systems

Feature	Indian system	International system
Grouping	first 3 digits from the right, then groups of **2**	groups of **3** digits
Separator	1,23,45,678	12,345,678
Names	thousand, lakh, crore	thousand, million, billion
Same number	12,00,000 (twelve lakh)	1,200,000 (1.2 million)
Same number	1,00,00,000 (one crore)	10,000,000 (10 million)

FORMULA

1 lakh = 100 thousand; 1 million = 10 lakh; 1 crore = 10 million; 1 billion = 100 crore.

Reading a number. 5,06,04,030 = **five crore six lakh four thousand thirty**. To write in the international style, regroup in threes: 50,604,030 (fifty million six hundred four thousand thirty).

1.3 Comparing and Operating on Large Numbers

- **Compare** by the number of digits first, then digit by digit from the left.
- **Add and subtract** by lining up place values: 3,45,000 + 2,67,500 = **6,12,500**.
- 7,05,000 is 10 times as large as 70,500 (7,05,000 ÷ 70,500 = 10).
- Population example: 75,000 in 2011 and 1,06,000 in 2024 gives an increase of **31,000**, about 41% of the 2011 figure.
- Always keep units: 'a tank holds 2,50,000 litres; pumps fill 5,000 litres/hour' means 2,50,000 ÷ 5,000 = **50 hours**.

1.4 Rounding and Estimation

DEFINITION

To **round** a number to a place, look at the digit to its right: 5 or more means round **up**, less than 5 means round **down**. An **estimate** uses rounded numbers to get a quick, approximate answer.

Number	Rounded to	Result
2,48,750	nearest thousand	2,49,000
3,84,400 km (Earth to Moon)	nearest ten thousand	3,80,000 km
4,980 × 302	estimate: 5,000 × 300	15,00,000 (exact 15,03,960)

Use estimation to check if an exact answer is **reasonable**: if your product of 4,980 × 302 comes to 1,50,396, the estimate 15,00,000 shows the decimal place is wrong.

1.5 Large Numbers in Everyday Contexts

Question	Working	Answer
Seconds in one day	24 × 60 × 60	86,400
Minutes in a year of 365 days	365 × 24 × 60	5,25,600 ≈ 5,30,000 (nearest ten thousand)
Hours to fill 2,50,000 L at 5,000 L/h	2,50,000 ÷ 5,000	50 hours
Can 3 new rice varieties daily over 100 years reach 1 lakh?	3 × 365 × 100	1,09,500, more than 1 lakh, so yes (if every rice is a new variety)

IMPORTANT

In estimation, the answer is a **good guess**, not the exact value. Always say what unit you used and to which place you rounded.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Write 5,06,04,030 in words.	Five crore six lakh four thousand thirty.
Largest 6-digit number? What after adding 1?	9,99,999; adding 1 gives 10,00,000.

Likely question	Model answer
Express 12,00,000 in the international system.	1,200,000 = 1.2 million.
How many seconds in one day?	86,400.
Round 2,48,750 to the nearest thousand.	2,49,000.
Find 3,45,000 + 2,67,500.	6,12,500.
How many lakhs make a crore?	100 lakhs.
1 billion equals how many crores?	100 crores.
Estimate 4,980 × 302.	About 5,000 × 300 = 15,00,000.
Compare 7,05,000 and 70,500.	The first is 10 times the second.

★ KEY TAKEAWAYS

- **1 lakh = 10^5; 1 crore = 10^7; 1 million = 10 lakh; 1 billion = 100 crore.**
- Indian grouping: 3 digits, then **2-2**; international: **3-3**.
- Compare numbers by digit count first, then from the left.
- **Round** using the digit to the right; **estimate** to check reasonableness.
- Convert times and quantities carefully: 1 day = **86,400 seconds**.

CHAPTER 02 · MATHS (PART 1)

Arithmetic Expressions

An arithmetic expression joins numbers with operations such as +, −, ×, ÷. This chapter shows how the same symbols can mean different things depending on brackets and order, how to write word statements as expressions, and how properties like the distributive law give quick ways to calculate.

Big Questions

- In what order do we evaluate an expression with several operations?
- How do brackets change the value and how do we write word statements as expressions?
- Which properties let us simplify or compare expressions without calculating everything?

2.1 What Is an Arithmetic Expression?

DEFINITION

An **arithmetic expression** is a combination of numbers and operation signs that has a **value**. An **equation** says that two expressions are equal (it has an = sign). 8 + 4 × 3 is an expression; 8 + 4 × 3 = 20 is an equation.

2.2 Order of Operations

Without a rule, 8 + 4 × 3 could be 36 or 20. The agreed order is:

Priority	Operation	Example
1	**B**rackets (innermost first)	(8 + 4) × 3 = 36
2	**O**rders / powers (squares, cubes)	$2 + 3^2 = 11$
3	**D**ivision and **M**ultiplication, left to right	18 ÷ 6 × 2 = 6
4	**A**ddition and **S**ubtraction, left to right	20 − 5 + 3 = 18

Example. 15 − 3 × 4 + 18 ÷ 6 = 15 − 12 + 3 = **6**, while (15 − 3) × 4 + 18 ÷ 6 = 48 + 3 = **51**. Likewise 50 − 10 + 5 = 45 but 50 − (10 + 5) = **35**: brackets change the value.

IMPORTANT

Multiplication and division have **equal rank**; so do addition and subtraction. Work **left to right** inside each rank. 24 ÷ 4 × 2 is **12**, not 3.

2.3 Putting Brackets to Use

- Insert brackets to get a target: 8 + 4 × 3 − 2 = 18, but (8 + 4) × 3 − 2 = **34**.
- In 4 + 5 × 6 − 2: (4 + 5) × 6 − 2 = **52**; 4 + 5 × (6 − 2) = 24.
- A **missing number** puzzle 7 × __ + 3 = 59 is solved backwards: 59 − 3 = 56, 56 ÷ 7 = **8**.
- ‘3 × (a number + 5) = 36’: number + 5 = 12, so the number is **7**.

2.4 Writing Word Statements as Expressions

Statement	Expression	Value
The difference of 50 and 18, divided by 8	(50 − 18) ÷ 8	4
Twice the sum of 7 and 5	2 × (7 + 5)	24
7 less than the product of 6 and 9	6 × 9 − 7	47
Half of the sum of 20 and 14	(20 + 14) ÷ 2	17

2.5 Properties That Save Work

Property	Statement	Use
Commutative	a + b = b + a; a × b = b × a	reorder to make friendly numbers: 25 × 7 × 4 = 25 × 4 × 7 = 700
Associative	(a + b) + c = a + (b + c); same for ×	regroup: 38 + 47 + 62 = 38 + 62 + 47
Distributive	**a × (b + c) = a × b + a × c**; a × (b − c) = a × b − a × c	35 × 98 + 35 × 2 = 35 × 100 = **3,500**
Subtraction and division	not commutative or associative	8 − 3 ≠ 3 − 8

Comparing without calculating. 13 × 15 = 195 and 14 × 14 = 196: the square is larger. In general $(n - 1)(n + 1) = n^2 - 1$, one less than n^2. Rewrite 36 + 24 as 12 × (3 + 2) = **60** by taking out the common factor 12.

> **NOTE**
> Writing the same number in different expression forms (e.g. 36 = 4 × 9 = 6 × 6 = 40 − 4) builds the habit of seeing structure, which is the foundation of **algebra** in the next chapter.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is an arithmetic expression? Example.	A combination of numbers and operations with a value, e.g. 8 + 4 × 3.
State the order of operations.	Brackets, powers, division and multiplication (left to right), addition and subtraction (left to right).
Insert brackets so that 8 + 4 × 3 − 2 = 34.	(8 + 4) × 3 − 2 = 34.
Evaluate 15 − 3 × 4 + 18 ÷ 6.	15 − 12 + 3 = 6.
Use the distributive property for 35 × 98 + 35 × 2.	35 × (98 + 2) = 3,500.
Find the missing number: 7 × __ + 3 = 59.	8.
Write ‘the difference of 50 and 18, divided by 8’ and evaluate.	(50 − 18) ÷ 8 = 4.
Compare 13 × 15 and 14 × 14.	195 < 196.

Likely question	Model answer
Is subtraction commutative?	No; 8 − 3 ≠ 3 − 8.
Why does 50 − 10 + 5 differ from 50 − (10 + 5)?	Brackets change the order: 45 versus 35.

★ KEY TAKEAWAYS

- Follow **BODMAS**: brackets, orders, division/multiplication, addition/subtraction; left to right within a rank.
- **Brackets** can change the value of an expression.
- **Distributive law:** a(b + c) = ab + ac: a quick way to calculate.
- Work **backwards** to find a missing number in an equation.
- Subtraction and division are **not** commutative or associative.

CHAPTER 03 · MATHS (PART 1)

A Peek Beyond the Point

Whole numbers cannot measure everything: a pencil is not exactly 14 cm, a bag of rice not exactly 3 kg. Decimals continue the place-value system to the right of the decimal point. This chapter covers place value, comparing and ordering, units of measure, rounding and the four operations on decimals.

Big Questions

- How does place value continue after the decimal point?
- How do we compare, order and round decimals?
- How do we add, subtract, multiply and divide decimals and convert units?

3.1 Place Value after the Point

Each place is **one-tenth** of the place to its left. After the ones place come **tenths (1/10), hundredths (1/100), thousandths (1/1000)**.

Digit in 34.567	Place	Value
3	tens	30
4	ones	4
5	tenths	0.5 = 5/10
6	hundredths	0.06 = 6/100
7	thousandths	0.007 = 7/1000

FORMULA

Expanded form: 3.206 = 3 + 2/10 + 0/100 + 6/1000. **Fraction form:** 0.45 = 45/100 = 9/20. 4 tenths + 5 hundredths = 0.45.

- Adding zeros at the **end** of a decimal does not change it: 12.3 = 12.30 = 12.300.
- A decimal point **separates** the whole part from the fractional part; money (₹12.50) and measures (3.25 m) use decimals.

3.2 Comparing, Ordering and Decimals between Decimals

Step 1
Compare the **whole parts**; the bigger whole part gives the bigger number.

Step 2
If they are equal, compare tenths, then hundredths, then thousandths.

Step 3
Add zeros if needed so both numbers have the same number of decimal places.

Example. 6.707 > 6.7 > 6.07 > 6.007 (descending). Also 0.5 > 0.45 and 3.07 < 3.7. Between any two decimals there are **infinitely many** more: between 1.4 and 1.5 we have 1.45, 1.41, 1.499, …

3.3 Units of Measurement and Conversions

Conversion	Rule	Example
cm → m	divide by 100	7 cm = 0.07 m; 3 m 7 cm = 3.07 m
mm → cm	divide by 10	45 mm = 4.5 cm
m → km	divide by 1000	2 km 50 m = 2.05 km
g → kg	divide by 1000	3 kg 450 g = 3.45 kg
₹ and paise	divide paise by 100	₹5 and 75 paise = ₹5.75

3.4 Rounding Decimals

Look at the digit **just after** the place you want to keep: 5 or more rounds up, less than 5 leaves the digit as it is. 4.678 to one decimal place is **4.7**; to two places **4.68**; to the nearest whole number **5**.

3.5 Operations on Decimals

Operation	Method	Example
Add	line up the decimal points, add as whole numbers	12.45 + 7.8 = 12.45 + 7.80 = **20.25**
Subtract	line up the points, borrow as needed	10 − 3.76 = 10.00 − 3.76 = **6.24**
Multiply by 10, 100, 1000	move the point **right** 1, 2, 3 places	3.456 × 100 = **345.6**
Divide by 10, 100, 1000	move the point **left** 1, 2, 3 places	47.5 ÷ 10 = **4.75**
Multiply two decimals	multiply as whole numbers, then place the point so that the decimal places = the total in both factors	4.2 × 3.5: 42 × 35 = 1470, 2 places → **14.70**
Divide a decimal by a whole number	divide as usual; put the point in the quotient above the point in the dividend	7.5 ÷ 5 = **1.5**

Word problems. Rice 3 kg 450 g plus wheat 2 kg 75 g: 3.45 + 2.075 = **5.525 kg**. Temperature rise from 36.4 °C to 38.9 °C is **2.5 °C**. A 4.2 m by 3.5 m rectangle has area **14.7 m^2** and perimeter 2 × (4.2 + 3.5) = **15.4 m**. If two decimals add to 8.5 and one is 3.275, the other is 8.5 − 3.275 = **5.225**.

> **IMPORTANT**
> Do not align decimals from the right like whole numbers: **8.5 + 3.275** needs 8.500 + 3.275. In subtraction, write extra zeros (10 = 10.00) before borrowing.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Write 4 tenths and 5 hundredths as a decimal.	0.45.
Expanded form of 3.206?	3 + 2/10 + 0/100 + 6/1000.

Likely question	Model answer
Find 10 − 3.76.	6.24.
Divide 47.5 by 10.	4.75.
Round 4.678 to one decimal place.	4.7.
Find a decimal between 1.4 and 1.5.	1.45 (any valid answer).
Arrange 6.07, 6.7, 6.007, 6.707 in descending order.	6.707, 6.7, 6.07, 6.007.
Write 3 m 7 cm in metres.	3.07 m.
Find 12.45 + 7.8.	20.25.
Why is 12.3 equal to 12.30?	A zero at the end of the decimal part does not change its value.

★ KEY TAKEAWAYS

- Places after the point: **tenths, hundredths, thousandths**, each 1/10 of the previous.
- Compare **whole part first**, then tenths, hundredths, …; end zeros do not matter.
- Between two decimals there are **infinitely many** decimals.
- **Line up the points** to add and subtract; multiply by 10, 100, 1000 by moving the point right.
- Convert units: cm → m ÷100, g → kg ÷1000, paise → ₹ ÷100.

CHAPTER 04 · MATHS (PART 1)

Expressions Using Letter-Numbers

A letter can stand for a number that is unknown or that changes. Such letter-numbers let us write rules once and use them for every case. This chapter shows how to form expressions, name their parts, simplify them by combining like terms, use them to describe patterns and substitute values.

Big Questions

- Why do we use letters for numbers and how do we form expressions?
- What are terms, coefficients, constants and like terms?
- How do expressions generalise patterns and how are they simplified and evaluated?

4.1 Letter-Numbers and Expressions

DEFINITION

A **letter-number (variable)** is a letter that stands for any number from a set of numbers. An **algebraic expression** is formed from numbers and letter-numbers joined by +, −, ×, ÷.

We write **3 × n as 3n**, **n × n as n^2** and omit the × sign between a number and a letter; 1 × n is simply n. The letter can be any symbol (x, y, p, n, …).

Situation	Expression	When the letter = 5
Cost of 12 notebooks at ₹p each	12p	₹60 (if p = 5)
12 notebooks and a ₹5 pen	12p + 5	65
Perimeter of a square of side s	4s	20
Area of a square of side s	s^2	25
Sum of three consecutive numbers starting at n	n + (n + 1) + (n + 2) = 3n + 3	18
₹x, spend ₹40 then get ₹15 back	x − 40 + 15 = x − 25	if x = 100, ₹75

4.2 Parts of an Expression

Word	Meaning	In 8k − 5
Term	a part added or subtracted	8k and −5
Coefficient	the number multiplying the letter	8
Constant	a term with no letter; its value never changes	−5
Like terms	terms with the same letter part	3x and 7x (but not 3x and 7y or $5x^2$)

REMEMBER

An expression with one term is a **monomial**, with two a **binomial**, with three a **trinomial**. Unlike terms cannot be combined: 3x + 4y stays as it is.

4.3 Generalising Patterns

The **nth term** formula describes a whole sequence. For 7, 12, 17, 22, … the difference is 5, so the nth term is **5n + 2** (check n = 1: 7). The 20th term is 5 × 20 + 2 = **102**.

Pattern	Rule (nth term)	Example
Even numbers 2, 4, 6, …	2n	10th = 20
Odd numbers 1, 3, 5, …	2n − 1	10th = 19
Multiples of 3	3n	8th = 24
Matchsticks for n squares in a row	3n + 1	5 squares: 16
Squares	n^2	7th = 49

4.4 Simplifying Expressions

Step 1 — Expand
Remove brackets using the distributive law: a(b + c) = ab + ac. Mind signs: −(x − 3) = −x + 3.

Step 2 — Group
Collect like terms together.

Step 3 — Combine
Add or subtract the coefficients, keep the letter part.

Expression	Working	Simplified
9y − 4y + 2y	(9 − 4 + 2)y	**7y**
3(x + 2) + 2(x − 1)	3x + 6 + 2x − 2	**5x + 4**
(7x + 5) − (3x − 2)	7x + 5 − 3x + 2	**4x + 7**
(4a + 3b) + (2a − 5b)	6a − 2b	**6a − 2b**
4(x + 3)	4x + 12	**4x + 12** (area of rectangle 4 by (x + 3))

4.5 Finding the Value of an Expression

Substitute the given number for the letter and use the order of operations. For $2x^2 − 3x + 1$ at x = 3: 2 × 9 − 9 + 1 = **10**. Finding the missing term: 5m + __ = 9m gives **4m**.

> **IMPORTANT**
> Substitute with brackets to avoid sign errors: for −3x at x = −2, write −3 × (−2) = **6**. Also, $3x^2$ means 3 × (x^2), not $(3x)^2$: for x = 2 these are 12 and 36.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is a letter-number? Why use it?	A letter that stands for an unknown or changing number; it lets one rule describe many cases.

Likely question	Model answer
Simplify 9y − 4y + 2y.	7y.
Coefficient and constant in 8k − 5?	Coefficient 8; constant −5.
Perimeter and area of a square of side s, for s = 5?	4s = 20; s^2 = 25.
Expression for sum of three consecutive numbers starting with n?	n + (n + 1) + (n + 2) = 3n + 3.
nth term of 7, 12, 17, 22, …? Find the 20th.	5n + 2; 102.
Simplify 3(x + 2) + 2(x − 1).	5x + 4.
Subtract 3x − 2 from 7x + 5.	4x + 7.
Which are like terms: 3x, 7y, 5x, $2x^2$?	3x and 5x.
Missing term: 5m + __ = 9m.	4m.

★ KEY TAKEAWAYS

- A **letter-number** stands for a changing or unknown number; 3 × n is written **3n**.
- An expression has **terms**; the number multiplying the letter is the **coefficient**; a lone number is a **constant**.
- Only **like terms** can be added or subtracted.
- The **nth term** formula (e.g. 5n + 2) generalises a pattern.
- **Expand** brackets, **collect** like terms, **substitute** carefully.